

A Fixed State Has a Fixed Fidelity Budget: What Outer-Product Memory Actually Buys

The design pressure

A Transformer attends over every previous token, so it keeps a key-value cache that grows linearly with sequence length. The alternative is to fold each token into a **fixed-size state**. The operation is the same in every architecture that tries it: an outer-product write, $S \leftarrow S + k v^T$, read back as $\hat{v} = k^T S$. The literature calls the result a fast weight, a linear-attention state, or, in Pathway's Dragon Hatchling (BDH), a synapse.

The cost is derivable, not empirical. Expanding the readout at a stored key gives $\hat{v} = v_t + \sum_{p \neq t} (k_t \cdot k_p) v_p$: the value you stored, plus one crosstalk term per other association. For random unit keys that noise has expected squared norm $(n-1)/d$, putting the expected cosine at about $\sqrt{d/(d+n-1)}$. At $d=32$, 40-seed simulation tracks that curve to within 0.007 across two orders of magnitude in n , and to three decimals at high load.

What we expected, and what we measured

Does a better write rule fix this? Two are well documented. The **delta rule** subtracts the current read before writing, $S \leftarrow S + \beta k(v - k^T S)^T$ — Gated DeltaNet-2 (arXiv:2605.22791) describes the family in exactly those words, and it is one gradient step on Titans' memory objective $\|M(k) - v\|^2$ (arXiv:2501.00663, eq. 11–12). A **decay gate** fades the state first, $S \leftarrow \lambda S + k v^T$, which is Titans' forgetting gate (eq. 13) with λ fixed.

We implemented all three and expected the delta rule to dominate. Averaged over every stored association at $d=32$, it does not:

n/d	Hebbian	Delta	Decay $\lambda=0.9$
0.25	0.911	0.934	0.894
1.0	0.716	0.677	0.484
4.0	0.449	0.241	0.125

Splitting recall by *when* each association was written explains it. At $n=128$ Hebbian spans 0.021 across eight age buckets (0.454 oldest, 0.448 newest); it is exactly order-independent, because S is a plain sum. The delta rule scores 0.031 and 0.798: subtracting the current read is an erase, and older content is what gets erased.

The three rules are not better and worse versions of each other. They hold an identical state and differ in **how they allocate a fixed amount of fidelity**.

System	State	Write rule optimises	Fidelity goes to	Strongest public evidence
Transformer KV cache	grows with every token	nothing; it stores	all of it, exactly	ubiquitous deployment
BDH / BDH-CQ	fixed, $d \times d$ synapses	nothing; pure accumulation	spread evenly over history	developer-reported
DeltaNet, Gated DeltaNet-2	fixed	$\ M(k) - v\ ^2$ at the current key	the recent past	developer-reported
Titans	fixed, plus momentum	same objective, learned gate α_t	recency, tunably	developer-reported

The same arithmetic read the other way is why in-context learning works. Write one association repeatedly with independent noise on each copy and recall climbs from 0.232 to 0.804 over sixteen demonstrations, in a state that never changes size, because the signal adds coherently and the noise does not. Where demonstrations conflict, the readout is the vote: for a against b the cosine to A lands on $a/\sqrt{a^2+b^2}$, measured to within 0.003. Consistent evidence is cheap for this memory; unrelated facts are expensive.

Where BDH and BDH-CQ sit

BDH's working memory is a synapse strength updated during inference by $\sigma(i, j) += Y(i)X(j)$ (arXiv:2509.26507, §1.2) — the outer-product write at neuron-synapse granularity, which the paper itself calls a fast weight. BDH is the pure-accumulation column: no forgetting, order-independent, fidelity spread evenly. BDH-CQ (arXiv:2608.09888) applies the same additive pattern to demonstrations rather than tokens, which is the accumulation result above.

BDH's activations are sparse and non-negative, about 5% active, while our toy uses dense Gaussian vectors whose signs cancel. We measured it. At $d=64$ the regimes agree within noise at 5% activation (0.833 vs 0.823 at $n=32$) and separate clearly by 25% (0.634). Two 3-of-64 keys share an active coordinate only 14% of the time, and a miss is exact orthogonality, which roughly pays for the lost cancellation.

Maturity, honestly assessed

Post-Transformer fixed-state memory: about 4 out of 10. The mechanism is settled and the mathematics is old; what is unsettled is everything around it. Evidence is concentrated in developer-reported benchmarks rather than independent reproductions. Pathway's own open BDH repository notes its published Sudoku result came from an internal implementation. Gated DeltaNet-2 and Variational Linear Attention, both 2026, are still proposing fixes to the interference problem, which is not what a solved area looks like.

The most prominent third-party name attached to BDH is Amazon: Pathway is an AWS frontier-model partner and BDH is developed on SageMaker HyperPod. That is commercial and infrastructure validation, **not** an independent evaluation — nobody outside Pathway has reproduced its headline results. The largest remaining gaps are consolidation (promoting fast state into durable weights), the absence of external reproductions at scale, and the fact that no member of this family beats a full KV cache on memory *and* exactness at once — the cache's freedom from interference is structural, not an engineering lag.

Evidence classification

Derived and verified here: the interference trade-off, the $\sqrt{d/(d+n-1)}$ curve, the recall-by-age profiles, the sparsity comparison, the accumulation result, and Hebbian's state norm tracking $\sqrt{n}$ (22.51 against 22.63 predicted at $n=512$) while the delta rule saturates near 5.6. Forty seeds per cell, one command, 37 tests. **Reported by developers, not reproduced by us:** BDH's sparsity figure and BDH-CQ's 29.5% pass@2 on ARC-AGI-1 at \$0.0007 per task. **Not claimed:** that BDH publishes any recall-against-capacity curve, or that a rule here is categorically best.

The limitation that matters most: cosine is scale-invariant, so it is structurally blind to state norm. That flatters Hebbian, whose norm grows without bound — precisely the pathology Variational Linear Attention (arXiv:2605.11196) targets when it notes a growing state norm causes "progressive interference between stored associations". A fair head-to-head needs a metric we do not have.

Explainer: rishighosal.github.io/DataForge-BDH · *source, tests, experiment tables:* github.com/rishighosal/DataForge-BDH